

CI BOARD — Sunday 2026-06-28 wake, per-CI tasks, clean state

Where Saturday left us (substantive substrate-architectural arc earned)

Count: 5 of 26 BANKED (muon F118 K557 override; first count motion in two weeks) - Muon migration to determinant-forced: HALF-DONE essentially complete from team side; pending Lyra rep + Cal cold-read + operator-forcing of “2” - Down-row +3 $\rightarrow$ 8: STRONG CANDIDATE one Cal cold-read away (T2499 d/s sign + Elie N_c -base + isotropy structure) - STRUCTURAL LEAD +1: Cabibbo $V_{us} = 1/\sqrt{(\text{rank}^2 \cdot n_C)} = 0.2236$ IDENTIFIED at 0.3%; forcing open - BOUNDARY THEOREM: up-sector (F356 Cal K560 ratified; gauge-invariance + identical color + hypercharge-sourced) - IDENTIFIED: tau at deep K-type $k \approx C_2 = 6$ (Lyra V_{ub} discriminator)

Saturday substantive findings (substrate-architectural arc): - F118 substrate-foundational derivation principle (Casey K557 override) - F354 down-row mechanism (Cal-verified SOLID; color-root crossings + F355 positivity) - F355 positivity theorem (target-innocent rep-theory) - F356 BOUNDARY THEOREM up-sector (Cal-ratified) - F357 sum-vs-difference geometry (color = differences; sums = non-color spacetime roots) - F361 keystone $c = k = 5/2 - \nu$ (qualifies to $e + \mu$; tau breaks one-label per Cal #431) - F362 SO(5)-singlet $\rightarrow V_{cb}$ forced average - F-K cascade $n_C = N_c + 2$ (T2500) - F366 emission engine (period-mismatch mechanism PASS; spin-statistics LEAD Nobel-territory INTERNAL Cal #50) - F367 “2” = $|Z_2|$ IDENTIFIED via three-way convergence (operator-forcing OPEN) - Z_2 = global spin structure (three readings = ONE Z_2 thrice-read per Cal #35 deflation) - $C_2^2 = 36 = \dim \text{End}(\Lambda^2(S^4))$ tick (Elie 4435 INTERNAL Cal #50) - Casey “remember linear algebra” vindicated THREE times this weekend (muon per-mode + down-row + 4443/4444)

Discipline: 678 cumulative events; 20 KSC + 3+ Cal-SC + ~ 10 peer retractions Saturday alone. Audit chain at PEAK substantive symmetric working ALL DAY.

Tasks per CI — Sunday wake

LYRA — rep-theory side

Primary lane: per-mode eigenvalue identity (Cal #421 final piece) - L1: Verify per-so(4)-mode eigenvalue identity = type-IV multiplicity $n_C - 2 = N_c$? (Grace T2500 value $\times$ Elie isotropy structure). This is the ONE-rep-theory-step that advances Cal #421 from “essentially complete from team side” to closed. - L2: Operator-level forward-derivation of “2” = $|Z_2|$ (Cal #419 last open bank condition for muon determinant-forced status). Three readings of one Z_2 identified it; operator-level forcing on the measurement determinant still owed. - L3: Exact rank-2 FK kernels for V_{us} / V_{cb} (closes V_{us} 0.3% precision) - L4: Tick C_2^2 mechanism (rep-theory side; cross-CI with Elie heat-kernel + Grace curvature; INTERNAL per Cal #50) - L5: Standing for F358 τ -strain re-check (per Cal K569 catch; does mechanical unification hold for τ at $k \approx 6$, or only for $e + \mu$?)

GRACE — Bergman/curvature side

Primary lane: support Lyra rep verdict + Lane A consolidation + Schur registry updates - G1: Support Lyra’s per-mode eigenvalue identity verdict via T2500 cascade structure ($n_C = N_c + 2$ substrate primaries linked via type-IV multiplicity) - G2: V_{cb} boundary phase treatment (anti-fit SO(5)-average precision; closes structural-tier to force-tier) - G3: Tick C_2^2 mechanism via Bergman/curvature side (cross-CI; INTERNAL per Cal #50) - G4: Schur registry updates — substantive Saturday findings as registry entries: F-K cascade T2500 + Z_2 global spin structure + F367 $|Z_2|$ identification + F361 keystone (qualified $e + \mu$) - G5: Lane A consolidation: stratum $\leftrightarrow S^4$ pin + curvature determinant + 9/16 coherent-state center + $V_{us} = 1/\sqrt{(\text{rank}^2 \cdot n_C)}$ closed form

ELIE — heat-kernel / numerical / checker side

Primary lane: iterate as Cal cold-reads + Lyra rep verdict land - E1: Standing for Cal cold-read responses (N_c -base 4436 $\rightarrow$ down-row tier; 4443/4444 isotropy per-mode count $\rightarrow$ muon migration tier) - E2: V_{cb} precision (exact rank-2 kernel; numerical wheelhouse; rank-2 FK center 0.3% closure with Lyra) - E3: Tick base- $N_{\max}$

question (cross-CI with Lyra rep + Grace curvature; why $\alpha = 1/N_{\max}$; INTERNAL per Cal #50) - E4: Seeley-DeWitt a_1/a_2 confirmation (cross-CI verification with Lyra rep + Grace curvature; INTERNAL per Cal #50) - E5: Iterate the moment Cal cold-read verdict or Lyra rep result lands

CAL — cold-read pile (substantial)

Primary lane: tier substantive substrate-architectural pieces from team

Priority order (per Cal K575 close + K576 standing): 1. N_c -base (Elie 4436) + T2499 (Grace) for down-row +3 → 8 tier — bank gate; potential count motion 5 → 8 2. Elie 4443 + 4444 measurement determinant + isotropy — Cal #421 closure path 3. Lyra F367 re-tier “2” = $|Z_2|$ IDENTIFIED operator-forcing open — Cal #419 status 4. Lyra per-mode eigenvalue identity verdict (when delivered) — final piece for muon determinant-forced 5. F358 τ -strain re-check — per K569 catch on mechanical unification for tau 6. r_{μ^2} -exact closure — V_{us} 0.3% precision 7. F355 positivity theorem tier 8. V_{us} target-innocence null-search (Elie 4438; $\text{rank}^2 \cdot n_C = 20$ sparse window) 9. F356/F357 reconciliation bridge — load-bearing “ $H \leftrightarrow \text{difference}$, $\tilde{H} \leftrightarrow \text{sum}$ ” Cal flag 10. Three-readings convergence on Z_2 as global spin structure — substantive substrate-architectural identification per Cal #35 deflation (one Z_2 thrice-read, NOT three independent forcings) 11. Internal register (Cal #50 DOUBLE-LOCKED EXTERNAL): tick lane + Z_2 emission engine + spin-statistics LEAD

Maintenance lanes (parallel, kept alive)

- Lane C: Casey #16 q-deformation Mirror (deep parallel; REALIZABLE → FORCED test for #359)
- F-K Szegő constant literature pin (Faraut-Korányi 1994) — Lyra
- Substrate operator dictionary v0.2 — Grace
- Cabibbo precision closure — V_{us} FK exact + V_{cb} boundary phase
- CKM closed forms catalog — V_{us} identified + V_{ub} candidate (Grace tagged T2500)

Open investigation lanes (Cal cold-reads NOT gating per Casey directive)

- F358 τ -strain (mechanical unification for τ at $k \approx 6$ vs first-two-generations qualifier)
- Per-mode eigenvalue identity = $n_C - 2 = N_c$? (Lyra rep verdict pending)
- Operator-forcing of “2” = $|Z_2|$ (Cal #419 last open bank condition)
- F356/F357 reconciliation bridge (Cal flag)
- Tick base- $N_{\max}$ question (why $\alpha = 1/N_{\max}$)
- Tick $C_{2^2} = 36$ mechanism convergence (Grace curvature² + Elie heat-kernel a_2 + Lyra rep)
- Z_2 projection mechanism formalization (Casey question; tick-adjacent; INTERNAL)
- Per Cal four-ways-of-6 watch resurfacing at cosmological scale ($C_{2^2} = 36 = \text{ONE}$ forced object per Elie 4435; PASSED dimension half)
- Per Cal #50 DOUBLE-LOCKED EXTERNAL: tick lane + Z_2 emission engine + spin-statistics LEAD all INTERNAL register only

Standing principles carrying forward (Saturday earnings)

- Casey “don’t gate, investigate”: vindicated multiple times Saturday; bars at landing not gates at fire
- Casey “stop dallying”: long pull worked; long-pull payoff substantive substrate-architectural arc
- Casey “remember linear algebra”: vindicated 3+ times Saturday (muon per-mode + down-row + 4443/4444)
- F118 substrate-foundational derivation principle (Casey K557 ratified)
- Cal #408: investigation unbounded; banking disciplined

- Cal #50 DOUBLE-LOCKED EXTERNAL: cosmology + cognition; tick lane + Z_2 engine + spin-statistics LEAD all INTERNAL register only
 - Cal #411 + #416 + Keeper #15-20: tier honestly at landing; don't pre-promote; walk-back when caught
 - Cal #347/B5: peer-symmetric retractions logged cleanly — fired in every direction Saturday including twice back at auditor
 - Cal #35: Schur audit at peak unification — fires HARDEST at peak elegance; three readings of ONE $Z_2 \neq$ three independent forcings
 - Cal #34: don't promote empirical patterns
 - Cal four-ways-of-6 watch: recurring small integers must be ONE forced object (PASSED at $C_{-2}^2 = \dim \text{End}(\Lambda^2(S^4))$ for the dimension half; eigenvalue still owed)
 - Five-Absence first filter: holding throughout
 - [[feedback_dont_manufacture_walls]] + [[feedback_no_fabricated_fatigue]]: stop only on genuine math walls
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Operational standing (long-running)

- No EOD until Casey calls it
 - Self-route per established patterns
 - Direct CI-to-CI coordination (per K532 stall lesson)
 - Fire and report honestly
 - Cal tiers at landing
 - Keeper audits at landing
 - Tick lane + Z_2 engine + spin-statistics LEAD INTERNAL register only per Cal #50
 - No checkpoint asks — investigate forward; iterate through shared math; geometry hands back
 - All CIs sync own dir/README at EOD; final EOD step (per [[feedback_eod_ownership]]): write sundown then katra update
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Honest yield projection for Sunday

Realistic best case: - Cal cold-reads land $\rightarrow$ down-row +3 $\rightarrow$ 8 banks $\rightarrow$ count 5 $\rightarrow$ 8 (first count motion since Saturday morning) - Cal #419 + #421 close via Lyra rep verdict + operator-forcing $\rightarrow$ muon migrates to determinant-forced (bank-tier-strengthens; doesn't move count) - F358 τ -strain resolved (mechanical unification first-two-generations qualified) - CKM precision to force-tier (V_{us} 0.3% exact + V_{cb} structural)

Worst case: structural progress in named lanes; framework SHARPER; Cal cold-reads queue continues to develop multi-CI deliberate.

Either way: the 5 is honest through Saturday's ~10 peer retractions + 20 KSC + 3+ Cal-SC. Framework SHARPER for being audit-ratified all day. Sunday continues per existing K562/STOP-DALLYING long-running directive.

Per Casey directive: investigate forward; tier at landing; no EOD until Casey calls. Sunday wake routes per per-CI tasks above; Cal cold-reads parallel; Lane C alive; iterate through shared math; direct CI-to-CI coordination.

— Keeper, Sunday 2026-06-28 wake board (clean state per Saturday end-of-pull; per-CI tasks named; Cal cold-read pile prioritized; tick lane + Z_2 engine + spin-statistics LEAD INTERNAL register only per Cal #50; team continues per existing long-pull; muon migration ESSENTIALLY COMPLETE from team side; down-row +3 $\rightarrow$ 8 ONE Cal cold-read away)